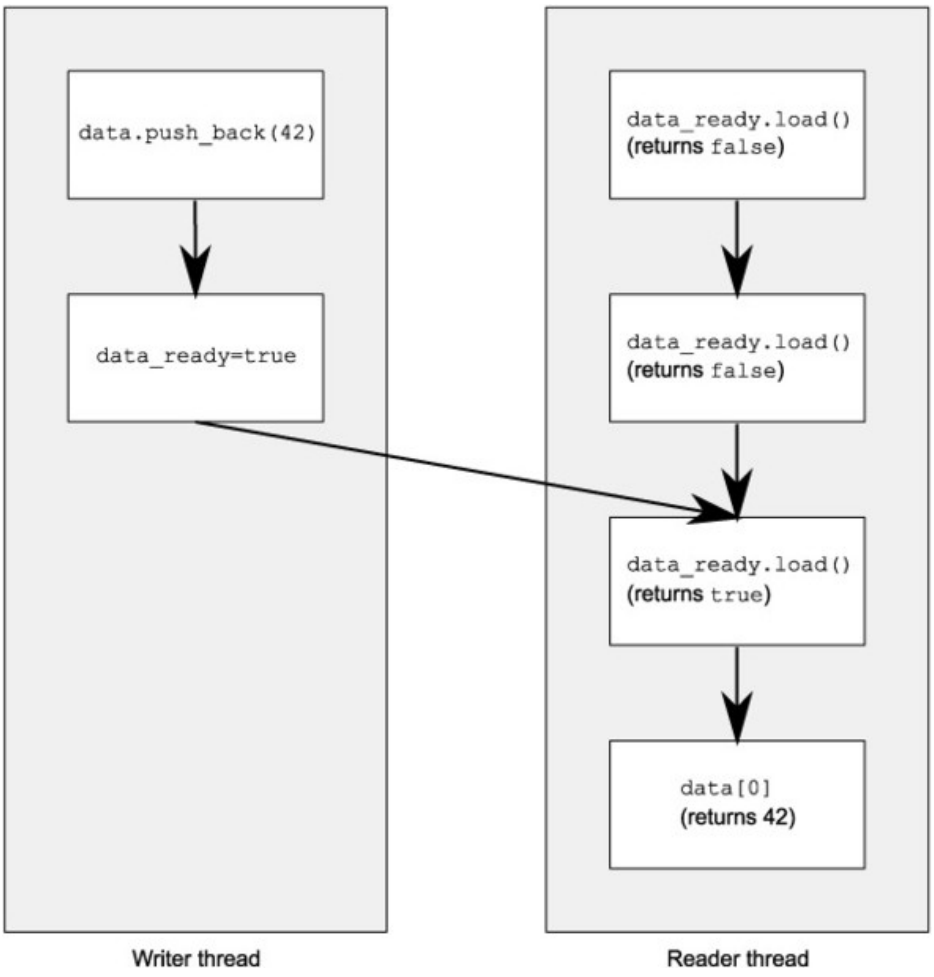


All this might seem fairly intuitive: of course the operation that writes a value happens before an operation that reads that value! With the default atomic operations, that's indeed true (which is why this is the default), but it does need spelling out: the atomic operations also have other options for the ordering requirements, which I'll come to shortly.

Now that you've seen happens-before and synchronizes-with in action, it's time to look at what they really mean. I'll start with synchronizes-with.

5.3.1. The synchronizes-with relationship

The synchronizes-with relationship is something that you can get only between operations on atomic types. Operations on a data structure (such as locking a mutex) might provide this relationship if the data structure contains atomic types and the operations on that data structure perform the appropriate atomic operations internally, but fundamentally it comes only from operations on atomic types.

The basic idea is this: a suitably tagged atomic write operation `W` on a variable `x` synchronizes-with a suitably tagged atomic read operation on `x` that reads the value stored by either that write (`W`), or a subsequent atomic write operation on `x` by the same thread that performed the initial